My Little



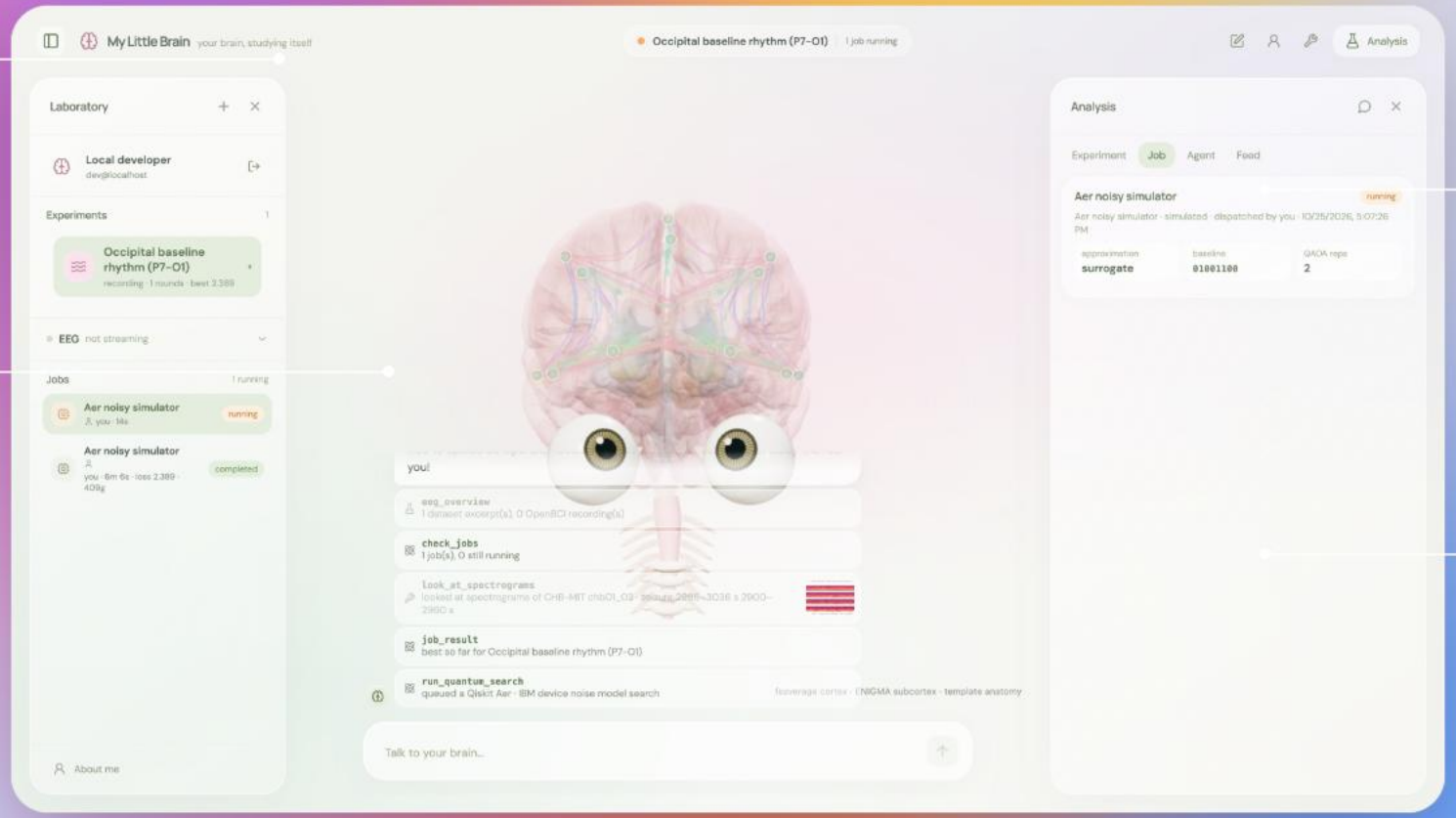
Brain

It knows you

your notes · your live EEG · your questions

it watches your live EEG

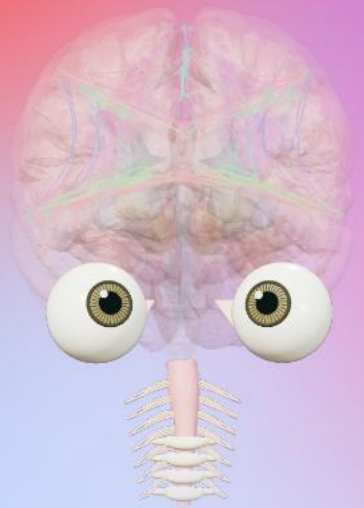
you chat with your brain



it designs quantum circuits

results land here

It studies itself



IONQ
IonQ · trapped ions

IBM
IBM · heavy-hex

qBraid
qBraid · Rigetti/IQM/AQT

Extropic · THRML

Gemini · vision + agent

OpenBCI · your EEG

48 / 256
simulations to recover a hidden brain state

42 gates
a circuit Gemini wrote itself (from 106)

52 vs 454
gates on IonQ vs IBM, same problem

τ 16→55 ms
seizure windows point to slower synapses

model hypotheses, not diagnoses · no quantum-advantage claim

Behind the scenes

YOU

OpenBCI headset

Cyton · 8 ch · 250 Hz
read in Chrome via Web Serial
+ your About-me notes

THE APP

Chat + 3D brain

React · three.js
real fsaverage anatomy
eyes · blinks · moods

GOOGLE CLOUD RUN

Gemini 4 Argon agent

chat · spectrogram vision
designs + revises circuits
FastAPI · one service

Cloud Tasks · Firestore · Cloud Storage
Secret Manager · Google Sign-In

GATE MODEL

QAOA

IonQ · IBM Quantum
qBraid · Qiskit Aer

THERMODYNAMIC

Extropic THRML

block-Gibbs on the Ising

CHECK

Forward sims

Wilson–Cowan · top-K
validated, never trusted

4 brain parameters × 2 bits = 8 qubits = 256 hypotheses · nonlinear loss → QUBO surrogate → exact Ising map

The team



Jacob Valdez

AI + robotics engineer
SuperCognition Labs · CommandAGI



Hanah Rahman

Student Researcher, Berkeley Lab
quantum applications · quantum ML



Rohit Korrapolu

Applied AI researcher
i14labs